MA3632 Lecture 6 – Linear and Polynomial Regression

In Lecture 5 we established the ERM framework: supervised learning is the problem of choosing a function $\hat{f}$ from a hypothesis class $\mathcal{H}$ to minimise the empirical risk under a chosen loss function. The k-nearest neighbour algorithm instantiated this framework with a non-parametric hypothesis class, making no assumptions about the form of f^*. In this lecture we take the opposite approach and impose explicit parametric structure.

Linear regression is ERM with the hypothesis class of affine functions and squared loss. This combination has a closed-form solution, a transparent geometric interpretation, and a direct connection to the singular value decomposition developed in Lecture 4. We derive the solution, study its geometric meaning, and then extend it in two directions: polynomial regression, which expands the hypothesis class while retaining the same solution machinery, and regularisation, which shrinks the solution towards zero to control variance. The classification analogue – logistic regression – is treated in Lecture 7.

1 Simple Linear Regression

We begin with the case $p = 1$: a single feature $x \in \mathbb{R}$ and a continuous target $y \in \mathbb{R}$. The hypothesis class is

$$\mathcal{H}_{\text{lin}} \;=\; \{f(x) = w_1 x + w_0 : w_0, w_1 \in \mathbb{R}\},$$

the set of all affine functions of x. Under squared loss, the empirical risk is

$$\hat{R}(w_0, w_1) \;=\; \frac{1}{n}\sum_{i=1}^{n}(y_i - w_1 x_i - w_0)^2.$$

Provided not every x_i takes the same value – that is, provided $\sum_{i=1}^{n}(x_i - \bar{x})^2 > 0$ – this is a smooth, strictly convex function of (w_0, w_1), so it has a unique global minimum obtained by setting both partial derivatives to zero. (If all x_i are identical, the slope is not identifiable: any line through the common point $(\bar{x}, \bar{y})$ with that x-value fits the data equally well in the w_1 direction, and $\hat{R}$ is only convex, not strictly so.)

Proposition 1.1 (Ordinary Least Squares, simple case)**.** *Suppose $\sum_{i=1}^{n}(x_i - \bar{x})^2 > 0$. Then the minimisers of $\hat{R}(w_0, w_1)$ are*

$$\hat{w}_1 \;=\; \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \qquad \hat{w}_0 \;=\; \bar{y} - \hat{w}_1\,\bar{x},$$

where $\bar{x} = n^{-1}\sum x_i$ and $\bar{y} = n^{-1}\sum y_i$.

Proof. Set $\partial\hat{R}/\partial w_0 = 0$ to obtain $\hat{w}_0 = \bar{y} - \hat{w}_1\bar{x}$. Substitute into $\partial\hat{R}/\partial w_1 = 0$ and simplify. The resulting equation for $\hat{w}_1$ is the ratio stated, which is well defined precisely because the denominator $\sum_i (x_i - \bar{x})^2$ is assumed positive. □

Remark 1.2. The formula for $\hat{w}_1$ is the sample covariance of x and y divided by the sample variance of x. It equals the sample correlation coefficient r_{xy} scaled by the ratio of standard deviations: $\hat{w}_1 = r_{xy}\, s_y/s_x$. This makes explicit that the slope depends on both the strength of the linear relationship and the relative spread of the variables.

The fitted line passes through the point $(\bar{x}, \bar{y})$ regardless of the data. The residuals $e_i = y_i - \hat{w}_1 x_i - \hat{w}_0$ satisfy $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$ – the residuals are orthogonal to the constant vector and to the vector of predictor values. This orthogonality condition is not merely algebraic; it is the key to the geometric interpretation developed in the next section.

Example 1.3. Suppose we have five observations: $(x, y) \in \{(1, 2), (2, 3), (3, 4), (4, 4), (5, 6)\}$. Then $\bar{x} = 3$, $\bar{y} = 3.8$, $\sum(x_i - \bar{x})(y_i - \bar{y}) = (-2)(-1.8) + (-1)(-0.8) + (0)(0.2) + (1)(0.2) + (2)(2.2) = 3.6 + 0.8 + 0 + 0.2 + 4.4 = 9$, and $\sum(x_i - \bar{x})^2 = 4 + 1 + 0 + 1 + 4 = 10$. So $\hat{w}_1 = 0.9$ and $\hat{w}_0 = 3.8 - 0.9 \times 3 = 1.1$. The fitted line is $\hat{f}(x) = 0.9x + 1.1$.

2 Multiple Linear Regression and the Normal Equations

When $p > 1$, it is convenient to absorb the intercept into the parameter vector by appending a column of ones to the design matrix. Define

$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix} \in \mathbb{R}^{n\times(p+1)}, \qquad \mathbf{w} = (w_0, w_1, \ldots, w_p)^\top \in \mathbb{R}^{p+1}, \qquad \mathbf{y} = (y_1, \ldots, y_n)^\top.$$

The empirical risk under squared loss is then

$$\hat{R}(\mathbf{w}) = \frac{1}{n}\|\mathbf{y} - X\mathbf{w}\|^2.$$

Theorem 2.1 (Normal Equations)**.** *If $X^\top X$ is invertible, the unique minimiser of $\hat{R}(\mathbf{w})$ is*

$$\hat{\mathbf{w}} = (X^\top X)^{-1} X^\top \mathbf{y}.$$

Proof. Expanding, $\hat{R}(\mathbf{w}) = n^{-1}(\mathbf{y}^\top\mathbf{y} - 2\mathbf{w}^\top X^\top \mathbf{y} + \mathbf{w}^\top X^\top X\mathbf{w})$. Since $X^\top X$ is positive definite (under the invertibility assumption), $\hat{R}$ is strictly convex. Setting the gradient $\nabla_{\mathbf{w}}\hat{R} = n^{-1}(-2X^\top\mathbf{y} + 2X^\top X\mathbf{w}) = \mathbf{0}$ gives the normal equations $X^\top X\,\hat{\mathbf{w}} = X^\top\mathbf{y}$, with unique solution $(X^\top X)^{-1}X^\top\mathbf{y}$. □

Remark 2.2. $X^\top X$ is invertible if and only if the columns of X are linearly independent, i.e. no predictor is an exact linear combination of the others. When predictors are nearly collinear, $X^\top X$ is nearly singular and $\hat{\mathbf{w}}$ has very high variance even if the matrix is technically invertible. This is the phenomenon of *multicollinearity*, and it is one of the main motivations for regularisation, treated in Section 4.

2.1 Geometric interpretation

The vector of fitted values $\hat{\mathbf{y}} = X\hat{\mathbf{w}}$ lies in the *column space* of X,

$$\mathrm{col}(X) = \{X\mathbf{w} : \mathbf{w} \in \mathbb{R}^{p+1}\} \subseteq \mathbb{R}^n.$$

Minimising $\|\mathbf{y} - X\mathbf{w}\|^2$ over $\mathbf{w}$ is equivalent to finding the point in $\mathrm{col}(X)$ closest to $\mathbf{y}$ in Euclidean distance. That closest point is the *orthogonal projection* of $\mathbf{y}$ onto $\mathrm{col}(X)$.

Proposition 2.3. *The fitted values $\hat{\mathbf{y}} = X(X^\top X)^{-1}X^\top\mathbf{y} =: H\mathbf{y}$ are the orthogonal projection of $\mathbf{y}$ onto* $\mathrm{col}(X)$*. The matrix $H = X(X^\top X)^{-1}X^\top$ is called the* hat matrix *and satisfies $H^2 = H$ (idempotent) and $H^\top = H$ (symmetric). The residual vector $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (I - H)\mathbf{y}$ is orthogonal to every column of X: $X^\top\mathbf{e} = \mathbf{0}$.*

Proof. $H^2 = X(X^\top X)^{-1}X^\top X(X^\top X)^{-1}X^\top = X(X^\top X)^{-1}X^\top = H$. Symmetry is immediate. Finally, $X^\top \mathbf{e} = X^\top(I-H)\mathbf{y} = (X^\top - X^\top H)\mathbf{y} = (X^\top - X^\top)\mathbf{y} = \mathbf{0}$. □

This geometric picture – prediction as orthogonal projection – connects linear regression directly to the singular value decomposition developed from first principles in Lecture 4, rather than assuming it as prerequisite background. Suppose X has full column rank $p+1 \le n$, with thin (economy) SVD $X = U\Sigma V^\top$, where $U \in \mathbb{R}^{n\times(p+1)}$ has orthonormal columns, $\Sigma \in \mathbb{R}^{(p+1)\times(p+1)}$ is diagonal with strictly positive entries, and $V \in \mathbb{R}^{(p+1)\times(p+1)}$ is orthogonal. Then $H = UU^\top$ and $\hat{\mathbf{w}} = V\Sigma^{-1}U^\top\mathbf{y}$, which is the Moore–Penrose pseudoinverse applied to $\mathbf{y}$. (When X does not have full column rank, the normal equations of Theorem 2.1 no longer have a unique solution, and the pseudoinverse formula must be adapted to omit the zero singular values; this case does not arise under the invertibility assumption of Theorem 2.1.) Computing $\hat{\mathbf{w}}$ via the SVD is numerically more stable than inverting $X^\top X$ directly, and is what most software implementations do in practice.

2.2 Assumptions and the Gauss–Markov theorem

The OLS estimator $\hat{\mathbf{w}}$ has good properties under a set of assumptions on the data-generating process. Suppose the true model is $\mathbf{y} = X\mathbf{w}^* + \boldsymbol{\varepsilon}$, where $\mathbf{w}^*$ is the true parameter vector and $\boldsymbol{\varepsilon}$ is a noise vector satisfying $\mathbb{E}[\boldsymbol{\varepsilon} \mid X] = \mathbf{0}$ and $\mathrm{Cov}(\boldsymbol{\varepsilon} \mid X) = \sigma^2 I$.

Theorem 2.4 (Gauss–Markov)**.** *Under the assumptions above,* $\hat{\mathbf{w}}$ *is the* Best Linear Unbiased Estimator *(BLUE) of* $\mathbf{w}^*$*: among all estimators that are linear in* $\mathbf{y}$ *and unbiased,* $\hat{\mathbf{w}}$ *has the smallest variance for every linear combination* $\mathbf{c}^\top\mathbf{w}^*$*.*

The Gauss–Markov theorem justifies OLS when the noise is homoscedastic (constant variance) and uncorrelated. It does not require normality. However, "best among linear unbiased estimators" is a limited guarantee: a biased estimator – one that systematically over- or underestimates $\mathbf{w}^*$ – can have lower mean squared error than $\hat{\mathbf{w}}$ if its bias is small and its variance reduction is large. Regularised estimators exploit exactly this trade-off, accepting a controlled amount of bias in exchange for substantially reduced variance.

3 Polynomial Regression

Linear regression restricts $\hat{f}$ to be an affine function of the raw features. When the true relationship is curved, this is too restrictive. Polynomial regression extends the hypothesis class by including powers of x as additional features, while keeping the model linear in the parameters.

Definition 3.1. Given a single feature x, the **degree-d polynomial regression model** is

$$f(x;\mathbf{w}) \;=\; w_0 + w_1 x + w_2 x^2 + \cdots + w_d x^d \;=\; \boldsymbol{\phi}(x)^\top \mathbf{w},$$

where $\boldsymbol{\phi}(x) = (1, x, x^2, \ldots, x^d)^\top$ is the *feature map* and $\mathbf{w} \in \mathbb{R}^{d+1}$ is the parameter vector. The design matrix is

$$\Phi \;=\; \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^d \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^d \end{pmatrix},$$

and the OLS estimate, provided Φ has full column rank, is $\hat{\mathbf{w}} = (\Phi^\top\Phi)^{-1}\Phi^\top\mathbf{y}$, identical in form to the linear case.

The key observation is that the model is *linear in the parameters* $\mathbf{w}$, even though it is nonlinear in x. This means the normal equations apply without modification; the only change is the design matrix. The same remark applies to any feature map $\boldsymbol{\phi}$: interactions, logarithms, trigonometric functions, and the polynomial features from Week 3 all give rise to linear-in-parameters models that can be fitted by OLS.

3.1 Overfitting and the bias–variance trade-off

The degree d is a hyperparameter that controls the complexity of $\hat{f}$. For $d = 1$ we recover linear regression. As d increases, the hypothesis class grows and the training error $\hat{R}$ falls monotonically – a degree-d polynomial can always achieve at least as low a training error as any degree less than d. When $d = n - 1$ and the x_i are distinct, a polynomial can interpolate all n training points exactly and the training error is zero.

Generalisation error tells a different story. A high-degree polynomial fitted to a moderate sample can oscillate strongly between training points, producing large prediction errors on new data. This is the overfitting regime: low bias (the polynomial can approximate many functions), high variance (it is extremely sensitive to the particular training sample). A low-degree polynomial may fit the training data poorly – high bias – but its predictions are stable. In practice, the degree is typically selected by cross-validation to balance these two contributions.

Example 3.2. Consider fitting polynomials of degree $d \in \{1, 3, 9\}$ to $n = 15$ points generated from $y = \cos(\pi x/2) + \varepsilon$ on $[0, 2]$ with $\varepsilon \sim N(0, 0.1^2)$. The degree-1 fit is a straight line; it systematically underestimates the curvature of the cosine (high bias). The degree-3 fit captures the curve well. The degree-9 fit passes near all 15 points but oscillates violently at the boundaries, achieving near-zero training error but very high test error. This Runge-type oscillation near the edges of the data range is a characteristic failure mode of high-degree polynomial regression.

4 Regularisation

Regularisation addresses overfitting by adding a penalty on the size of the coefficients to the empirical risk, discouraging solutions with large coefficients. Large coefficients can be associated with models that are sensitive to noise, particularly in flexible or ill-conditioned settings, so penalising coefficient magnitude can reduce variance at the cost of a small increase in bias.

For this section it is convenient to separate the intercept from the remaining coefficients rather than absorbing it into X as in Section 2. Write the model as $f(\mathbf{x}) = w_0 + \mathbf{x}^\top \mathbf{w}$, where $w_0 \in \mathbb{R}$ is the intercept and $\mathbf{w} = (w_1, \ldots, w_p)^\top \in \mathbb{R}^p$ are the remaining coefficients, and let $X \in \mathbb{R}^{n \times p}$ denote the design matrix of the p predictors *without* an intercept column. As discussed further in Section 4.1 and Exercise 3, the intercept is not penalised: only $\mathbf{w}$ is shrunk towards zero.

4.1 Ridge regression

Definition 4.1. Ridge regression (also called ℓ_2 regularisation or Tikhonov regularisation) minimises the penalised empirical risk

$$\hat{R}_\lambda(w_0, \mathbf{w}) \;=\; \frac{1}{n}\|\mathbf{y} - w_0\mathbf{1} - X\mathbf{w}\|^2 + \lambda\|\mathbf{w}\|^2,$$

where $\lambda > 0$ is the *regularisation parameter* and the penalty applies only to $\mathbf{w}$, not to the intercept w_0.

Proposition 4.2. *Suppose the columns of X have been centred to have sample mean zero. Then the Ridge estimator has a unique closed-form solution for any $\lambda > 0$:*

$$\hat{w}_0 \;=\; \bar{y}, \qquad \hat{\mathbf{w}}_\lambda \;=\; (X^\top X + n\lambda I)^{-1} X^\top \mathbf{y}.$$

Proof. Since the columns of X are centred, $X^\top \mathbf{1} = \mathbf{0}$, so the intercept and slope terms decouple in the normal equations: differentiating $\hat{R}_\lambda$ with respect to w_0 and setting the result to zero gives $\hat{w}_0 = \bar{y}$ regardless of $\mathbf{w}$ (exactly as in ordinary OLS, since w_0 is unpenalised). Substituting $\hat{w}_0 = \bar{y}$, the penalised risk in $\mathbf{w}$ alone, $n^{-1}\|(\mathbf{y} - \bar{y}\mathbf{1}) - X\mathbf{w}\|^2 + \lambda\|\mathbf{w}\|^2$, is strictly convex, since $X^\top X + n\lambda I$ is positive definite regardless of whether $X^\top X$ is invertible. Setting its gradient to zero gives $(X^\top X + n\lambda I)\hat{\mathbf{w}}_\lambda = X^\top(\mathbf{y} - \bar{y}\mathbf{1}) = X^\top \mathbf{y}$, using $X^\top \mathbf{1} = \mathbf{0}$ again in the last step. □

The effect of the $n\lambda I$ term is to shift all eigenvalues of $X^\top X$ upward by $n\lambda$, making the matrix well-conditioned even when $X^\top X$ is singular or nearly so. Ridge therefore directly mitigates the instability caused by multicollinearity. In terms of the SVD $X = U\Sigma V^\top$, and using $U^\top \mathbf{1} = \mathbf{0}$ together with $\hat{w}_0 = \bar{y}$, the Ridge estimator can be written

$$\hat{\mathbf{w}}_\lambda \;=\; V \operatorname{diag}\!\left(\frac{\sigma_j}{\sigma_j^2 + n\lambda}\right) U^\top(\mathbf{y} - \bar{y}\mathbf{1}) \;=\; V \operatorname{diag}\!\left(\frac{\sigma_j}{\sigma_j^2 + n\lambda}\right) U^\top \mathbf{y},$$

where σ_j are the singular values of X; the two expressions agree because $U^\top \mathbf{1} = \mathbf{0}$ removes any contribution from $\bar{y}\mathbf{1}$. Small singular values – corresponding to directions in the feature space that are poorly determined by the data – are shrunk more aggressively than large ones.

Remark 4.3. Ridge shrinks $\hat{\mathbf{w}}_\lambda$ towards zero as λ grows, and each coordinate $\sigma_j/(\sigma_j^2 + n\lambda)$ decreases monotonically to zero as $\lambda \to \infty$, but for finite λ the estimator does not generally produce sparse solutions: unlike the Lasso below, it does not systematically set coefficients exactly to zero. It performs *shrinkage* but not, in general, *selection.* In high-dimensional problems where many features are irrelevant, this is a limitation.

4.2 Lasso regression

Definition 4.4. The **Lasso** (Least Absolute Shrinkage and Selection Operator) minimises

$$\hat{R}_\lambda(w_0, \mathbf{w}) \;=\; \frac{1}{n}\|\mathbf{y} - w_0\mathbf{1} - X\mathbf{w}\|^2 + \lambda\|\mathbf{w}\|_1,$$

where $\|\mathbf{w}\|_1 = \sum_{j=1}^p |w_j|$ is the ℓ_1 norm of the (unpenalised intercept aside) coefficient vector $\mathbf{w} = (w_1, \ldots, w_p)^\top$.

Unlike Ridge, the Lasso penalty is not differentiable at zero, and there is no closed-form solution in general. The problem is a convex quadratic programme and is solved efficiently by coordinate descent or the LARS algorithm. The qualitative behaviour, however, is markedly different from Ridge: the ℓ_1 penalty has a tendency to drive some coefficients to exactly zero, performing *simultaneous shrinkage and variable selection.*

The geometric reason for this difference can be seen by restating the two estimators as constrained optimisation problems. Ridge solves $\min_{\mathbf{w}} \|\mathbf{y} - X\mathbf{w}\|^2$ subject to $\|\mathbf{w}\|^2 \le t$, for some t depending on λ. Lasso solves the same problem subject to $\|\mathbf{w}\|_1 \le t$. The ℓ_2 constraint set is a smooth ball; the ℓ_1 constraint set is a polytope with corners on the coordinate axes. As the unconstrained minimum is moved towards the constraint set, it is much more likely to first contact a corner of the ℓ_1 polytope than a smooth point of the ℓ_2 ball. Corners correspond to sparse solutions where some $w_j = 0$.

Remark 4.5. The choice between Ridge and Lasso depends on the assumed structure of f^*. If many features have small but nonzero effects, Ridge is usually better; it shrinks them all without discarding any. If only a few features are relevant and the rest are noise, Lasso is preferred because it produces an interpretable sparse model. In practice, the *elastic net*, which combines both penalties, $\lambda_1\|\mathbf{w}\|_1 + \lambda_2\|\mathbf{w}\|^2$, is often used when neither assumption is clearly correct.

4.3 Choosing λ by cross-validation

The regularisation parameter λ is a hyperparameter and is typically chosen by cross-validation, just as k was chosen in Lecture 5. One fits the model for a grid of λ values, computes the cross-validation error at each, and selects the value minimising (or within one standard error of) the minimum. In scikit-learn, `RidgeCV` and `LassoCV` implement this with efficient computation over a path of λ values rather than by fitting each value independently.

For polynomial regression, one can regularise in addition to choosing d: fit a high-degree polynomial but penalise the size of the coefficients. This often outperforms selecting d by cross-validation alone, since it allows a richer polynomial basis while controlling effective model complexity continuously through λ, rather than only through the coarser, discrete choice of degree.

5 Assessing Regression Models

Once a model is fitted, its performance must be assessed on held-out data using an appropriate metric. The most common metrics for regression are the following.

Definition 5.1. Let $\hat{y}_i$ be the prediction for the i-th observation in a test set of size m.

(i) **Mean squared error** (MSE): $\mathrm{MSE} = m^{-1}\sum_{i=1}^m (y_i - \hat{y}_i)^2$.

(ii) **Root mean squared error** (RMSE): $\mathrm{RMSE} = \sqrt{\mathrm{MSE}}$. RMSE is on the same scale as y and is therefore easier to interpret than MSE.

(iii) **Mean absolute error** (MAE): $\mathrm{MAE} = m^{-1}\sum_{i=1}^m |y_i - \hat{y}_i|$. MAE is less sensitive to large errors than RMSE.

(iv) **Coefficient of determination** (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^m (y_i - \hat{y}_i)^2}{\sum_{i=1}^m (y_i - \bar{y})^2},$$

where $\bar{y}$ is the mean of the test targets. R^2 is one minus the ratio of the model's squared error to that of the constant-mean baseline predictor; on the training set of a linear model this ratio has the further interpretation of one minus the proportion of variance in y left unexplained by $\hat{f}$. A value of $R^2 = 1$ indicates a perfect fit; $R^2 = 0$ means the model does no better than predicting the mean; $R^2 < 0$ means the model is worse than the mean predictor, which can and does happen on a held-out test set even for a model that fitted the training data reasonably well.

Remark 5.2. R^2 computed on the training set is not a reliable measure of generalisation: it increases monotonically with model complexity and equals one for any interpolating model. R^2 should be reported on a held-out test set, or via cross-validation, to be meaningful. Similarly, the "adjusted R^2" corrects for the number of parameters only in a linear model under Gaussian noise assumptions; it is not a substitute for proper out-of-sample evaluation.

6 Exercises

Exercise 6.1. Consider the following four observations: $(x_1, y_1) = (0, 1)$, $(x_2, y_2) = (1, 3)$, $(x_3, y_3) = (2, 2)$, $(x_4, y_4) = (3, 5)$.
Write down the design matrix X (with intercept column) and the target vector $\mathbf{y}$. Compute $X^\top X$ and $X^\top \mathbf{y}$, solve the normal equations, and state the OLS fitted line $\hat{f}(x) = \hat{w}_0 + \hat{w}_1 x$. Verify that the residuals sum to zero and are orthogonal to the predictor values.

Solution. The design matrix and target vector are

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \qquad \mathbf{y} = \begin{pmatrix} 1 \\ 3 \\ 2 \\ 5 \end{pmatrix}.$$

Then

$$X^\top X = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}, \qquad X^\top \mathbf{y} = \begin{pmatrix} 11 \\ 22 \end{pmatrix}.$$

The normal equations $X^\top X\,\hat{\mathbf{w}} = X^\top \mathbf{y}$ give

$$\begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix} \begin{pmatrix} \hat{w}_0 \\ \hat{w}_1 \end{pmatrix} = \begin{pmatrix} 11 \\ 22 \end{pmatrix}.$$

From the first equation, $\hat{w}_0 = (11 - 6\hat{w}_1)/4$. Substituting into the second: $6(11 - 6\hat{w}_1)/4 + 14\hat{w}_1 = 22$, so $16.5 - 9\hat{w}_1 + 14\hat{w}_1 = 22$, giving $5\hat{w}_1 = 5.5$ and $\hat{w}_1 = 1.1$. Then $\hat{w}_0 = (11 - 6.6)/4 = 1.1$.
The fitted line is $\hat{f}(x) = 1.1 + 1.1x$.
Fitted values: $\hat{y}_i = 1.1, 2.2, 3.3, 4.4$. Residuals: $e_i = 1 - 1.1, 3 - 2.2, 2 - 3.3, 5 - 4.4 = -0.1, 0.8, -1.3, 0.6$.
Sum of residuals: $-0.1 + 0.8 - 1.3 + 0.6 = 0$, confirming the orthogonality of the residuals to the constant vector.
Inner product with predictor values: $\sum x_i e_i = 0(-0.1) + 1(0.8) + 2(-1.3) + 3(0.6) = 0 + 0.8 - 2.6 + 1.8 = 0$, confirming orthogonality to the predictor values as well. □

Exercise 6.2. Let $X \in \mathbb{R}^{n \times (p+1)}$ have full column rank and let $\hat{\mathbf{w}} = (X^\top X)^{-1} X^\top \mathbf{y}$ be the OLS estimator under the model $\mathbf{y} = X\mathbf{w}^* + \boldsymbol{\varepsilon}$ with $\mathbb{E}[\boldsymbol{\varepsilon} \mid X] = \mathbf{0}$ and $\mathrm{Cov}(\boldsymbol{\varepsilon} \mid X) = \sigma^2 I$.
Show that $\hat{\mathbf{w}}$ is an unbiased estimator of $\mathbf{w}^*$, i.e. $\mathbb{E}[\hat{\mathbf{w}} \mid X] = \mathbf{w}^*$. Then compute $\mathrm{Cov}(\hat{\mathbf{w}} \mid X)$ and show that it equals $\sigma^2 (X^\top X)^{-1}$. Explain in words what this covariance matrix implies for the precision of individual coefficient estimates when two predictors are nearly collinear.

Solution. Write $\hat{\mathbf{w}} = (X^\top X)^{-1} X^\top \mathbf{y} = (X^\top X)^{-1} X^\top (X\mathbf{w}^* + \boldsymbol{\varepsilon}) = \mathbf{w}^* + (X^\top X)^{-1} X^\top \boldsymbol{\varepsilon}$. Taking expectations conditional on X and using $\mathbb{E}[\boldsymbol{\varepsilon} \mid X] = \mathbf{0}$:

$$\mathbb{E}[\hat{\mathbf{w}} \mid X] \;=\; \mathbf{w}^* + (X^\top X)^{-1} X^\top \mathbf{0} \;=\; \mathbf{w}^*.$$

For the covariance, let $A = (X^\top X)^{-1} X^\top$, which is a fixed matrix once we condition on X. Then $\hat{\mathbf{w}} - \mathbf{w}^* = A\boldsymbol{\varepsilon}$, so

$$\mathrm{Cov}(\hat{\mathbf{w}} \mid X) \;=\; A\,\mathrm{Cov}(\boldsymbol{\varepsilon} \mid X)\,A^\top \;=\; \sigma^2 A A^\top \;=\; \sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-\top} \;=\; \sigma^2 (X^\top X)^{-1}.$$

When two predictors x_j and x_k are nearly collinear, $X^\top X$ is nearly singular, so $(X^\top X)^{-1}$ can have large entries, including large diagonal entries corresponding to inflated variances of individual coefficient estimates $\hat{w}_j$. Near-collinearity therefore inflates these variances: small perturbations to the data – or to the noise $\boldsymbol{\varepsilon}$ – can shift the estimated coefficients dramatically, even if the fitted values $X\hat{\mathbf{w}}$ are stable. Ridge regression addresses this by adding $n\lambda I$ to $X^\top X$, which bounds the diagonal entries of the inverse and stabilises the coefficient estimates. □

Exercise 6.3. Suppose a Ridge regression model is fitted with regularisation parameter $\lambda > 0$ and the features have been standardised to zero mean and unit variance. Explain why the intercept is typically excluded from the Ridge penalty, i.e. why one minimises $n^{-1}\|\mathbf{y} - X\mathbf{w} - w_0\mathbf{1}\|^2 + \lambda\|\mathbf{w}\|^2$ rather than also penalising w_0^2. What is the OLS estimate of w_0 when features are centred, and does Ridge change this estimate?

Solution. The intercept w_0 captures the overall mean level of y and does not control the complexity of the fitted surface; penalising it would force the fitted function to pass near zero regardless of $\bar{y}$, which is not a useful constraint. Penalising the slope coefficients $\mathbf{w}$ is the intended effect of regularisation: it discourages the model from assigning unnecessarily large weights to individual features, which can make the fit more sensitive to noise.
When features are centred (zero mean), the design matrix X (columns for the predictors only) satisfies $X^\top\mathbf{1} = \mathbf{0}$, so the intercept and slope coefficients are orthogonal in the normal equations. The OLS estimate of the intercept is $\hat{w}_0 = \bar{y}$ regardless of the slope estimates. Since Ridge only penalises $\mathbf{w}$, the intercept equation is unchanged from OLS, and Ridge also gives $\hat{w}_0 = \bar{y}$, as derived in Proposition 4.2. Standardising the features before fitting Ridge (or Lasso) is therefore standard practice: it ensures the penalty treats all predictors on an equal footing and decouples the intercept from the regularised coefficients. □

Further Reading

Lindholm, A., Wahlström, N., Lindsten, F., and Schön, T. B. (2022). *Machine Learning: A First Course for Engineers and Scientists*. Cambridge University Press. Treats linear regression, the normal equations, and regularisation as part of a unified parametric modelling framework, at a level consistent with this lecture.

James, G., Witten, D., Hastie, T., Tibshirani, R., and Taylor, J. (2023). *An Introduction to Statistical Learning with Applications in Python*. Springer. Chapter 3 covers simple and multiple linear regression in detail, including the standard diagnostic plots used in the Week 6 workshop; Chapter 6 covers Ridge regression, the Lasso, and the geometric constraint-region argument used in Section 4.2.

Hoerl, A. E. and Kennard, R. W. (1970). Ridge regression: biased estimation for nonorthogonal problems. *Technometrics*, 12(1), 55–67. The original paper introducing Ridge regression as a remedy for multicollinearity.

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